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Undergraduate Thesis (Design)

Thesis Title	LongMedBench: Benchmarking Memory-Augmented Medical Agents f
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Class of year	Class of 2026
Major	Computer Engineering
Submission date	May 2026

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Acknowledgements

I would like to express my sincere gratitude to my supervisor, Professor Zuozhu Liu, for his guidance, patience, and insightful feedback throughout the course of this thesis project. His expertise in medical AI provided the intellectual foundation upon which this work was built.

I am grateful to my collaborators on the LongMedBench project, Yanzhen Cheng and Xiaocheng Zhang, whose work on the factual question-answering and temporal reasoning components provided essential context for the decision-making framework developed in this thesis. LongMedBench has been submitted to MICCAI 2026 and the rebuttal stage is completed recently.

Many thanks to the senior students of ZJUI-AI4H, especially Yuan Wang who provided us with computing resources, and Zhiting Fan who who provided concrete guidance on paper writing. The collaborative environment fostered productive discussions that sharpened the research questions and methodology.

I also acknowledge the maintainers of the MIMIC-IV database, whose commitment to open clinical data access makes research of this kind possible.

Finally, I thank my family for their unwavering support throughout my undergraduate studies, and the faculty of the Computer Engineering program at ZJUI and UIUC for providing a rigorous and inspiring academic environment.

Abstract

Clinical decision-making is inherently longitudinal — clinicians must synthesize evidence across repeated hospitalizations, diagnostic tests, and evolving treatment responses spanning months or years. However, current benchmarks for medical large language model (LLM) predominantly evaluate single-encounter behavior, focusing on model knowledge, tool-use accuracy and fact retrieval rather than the cross-visit temporal integration that real clinical practice demands. To address this gap, this thesis introduces the decision-making component of LongMedBench, a benchmark constructed from real-world Electronic Health Record (EHR) data (MIMIC-IV) that evaluates LLM agents on long-horizon clinical decision-making. We design a three-tier memory architecture operating at three granularities — Note Memory (visit-level summaries), Event Memory (fine-grained structured event sequences), and Contextual Memory (immediate visit context) — and systematically compare memory augmentation strategies including naive long-context, retrieval-augmented generation (RAG), and agent memory systems (Mem0). Through experiments on GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo, we find that decision-making accuracy across all models remains modest, and critically, that memory augmentation architectures provide no significant improvement over the naive long-context baseline. Decision accuracy depends primarily on the immediate clinical context, with historical information contributing negligible marginal value regardless of memory granularity or retrieval strategy. This central finding reveals a retrieval-reasoning gap: current architectures can locate relevant historical facts but cannot effectively integrate them into proactive clinical planning. We discuss the implications of this gap for medical AI agent design and propose directions for memory architectures that go beyond retrieval to support structured longitudinal reasoning.

Keywords: Medical AI Agents, Clinical Decision-Making, Electronic Health Records, Long-Horizon Reasoning, Retrieval-Augmented Generation, Agent Memory Systems

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Chapter 1. Introduction

1.1 The Long-Horizon Challenge in Clinical Decision-Making

Clinical practice is fundamentally longitudinal. Patients with chronic conditions accumulate medical histories spanning years and dozens of hospitalizations, and at each encounter, clinicians must synthesize evidence across prior diagnostic tests, imaging studies, medication adjustments, and evolving treatment responses [1]. This capacity for cross-visit temporal integration is central to clinical reasoning and represents a capability that medical AI must acquire to be clinically useful.

As large language models (LLMs) transition from static question-answering to autonomous medical agents, their ability to navigate longitudinal clinical trajectories has become a critical benchmark for clinical readiness. Yet existing evaluation benchmarks assess single-encounter behavior: tool-use accuracy in a simulated visit, diagnosis from a single symptom set, or fact retrieval from a single discharge summary. These evaluations, while valuable, do not measure the cross-visit temporal integration that real clinical practice demands.

This thesis presents the decision-making component of **LongMedBench**, a benchmark constructed from MIMIC-IV [2] EHR data that evaluates LLM agents on long-horizon clinical decision-making. Our work bridges two previously disconnected lines of research: the design of long-horizon interaction benchmarks from the general AI domain, and the evaluation of clinical reasoning from real-world EHR data. Through a three-tier memory architecture and systematic ablation of memory strategies, we investigate a central question: given long-context windows, RAG, and agent memory systems, how well can contemporary LLMs integrate longitudinal clinical history to inform next-step decisions?

1.2 Medical Benchmarks for Clinical AI

Medical AI evaluation has evolved from static QA to interactive agent settings, as illustrated in Table 1.1. Early benchmarks such as MedBench [3] and PromptCBLUE

Table 1.1 Comparison of medical benchmarks for clinical AI. *Legend:* EHR = real electronic health record data. Agent = interactive agent-based evaluation. Memory Study = systematic memory architecture ablation.

Benchmark	EHR	Multi-Visit	Agent	Decision-Making	Temporal	Memory Study	Year	Data Source
MedBench [3]							2023	Exams
EHRSQL [4]	✓						2022	MIMIC-IV
MedAlign [5]	✓			✓			2024	STARR
MedAgentBench [6]	✓		✓	✓			2025	STARR
AgentClinic [7]	✓		✓	✓			2025	Simulated
ER-REASON [8]	✓		✓	✓			2025	MIMIC-IV
FHIR-AgentBench [9]	✓		✓		✓		2025	MIMIC-IV
MedAgentGym [10]	✓		✓	✓			2025	Multiple
TIMER [11]	✓	✓	✓	✓	✓		2025	STARR
EHR-Bench [12]	✓	✓	✓	✓			2025	MIMIC-IV
LongMedBench	✓	✓	✓	✓	✓	✓	2026	MIMIC-IV

[13] tested LLMs' factual knowledge recall from medical licensing exams. Second-generation work introduced real clinical data: EHRSQL [4] pioneered text-to-SQL on structured EHR; FHIR-AgentBench [9] extended this to interoperable FHIR standards; MedAlign [5] provided clinician-generated instruction-following tasks grounded in EHR data.

Both categories above assess static model capabilities; however, real-world medical tasks demand interactive reasoning. A series of agent-based evaluation frameworks has been developed, wherein researchers construct simulated environments that enable LLMs to interactively complete complex tasks. MedAgentBench [6] introduced a virtual EHR environment with 300 clinician-written tasks; AgentClinic [7] added multimodal clinical encounters; ER-REASON [8] focused on emergency room reasoning with structured subtasks; and MedAgentGym [10] provided a scalable training environment for biomedical reasoning.

Despite this progress, a critical gap persists: all current medical benchmarks evaluate single-encounter behavior. Real clinical care spans dozens of visits over years, and a patient's current presentation cannot be understood in isolation from their clinical trajectory. Furthermore, no existing medical benchmark systematically compares memory architectures for AI agents in clinical decision-making.

1.3 Long-Horizon Benchmarks in the General Domain

In parallel with medical AI, a rich set of long-horizon benchmarks has emerged in the broader AI agent community, as shown in Table 1.2. Their task design principles inspire the construction of longitudinal clinical benchmarks, which include multi-turn interaction, cross-session memory, and progressive information revelation.

Long-Context Retrieval. RULER [14] systematically evaluates the effective context window of LLMs through synthetic tasks such as needle-in-a-haystack retrieval and variable tracking at controlled lengths up to 128K tokens. NarrativeQA [15] tests reading comprehension over full-length books and movie scripts, requiring models to answer questions that depend on information distributed across long documents. HotpotQA [16] requires reasoning across multiple documents to answer questions.

Multi-Hop and Multi-Turn Benchmarks. Beyond basic retrieval, LoCoMo [17] and LongMemEval [18] use simulated multi-session dialogues to evaluate whether models can recall and reason about information shared across temporally separated interactions. Both introduce the concept of multi-hop reasoning, which parallels the clinical need to connect evidence across visits.

Interactive Agent Benchmarks. TRIP-Bench [19] evaluates agents on executing long-horizon interactive tasks in real-world scenarios, where agents must plan, execute, and adapt across extended interactions. AMA-Bench [20] benchmarks long-horizon memory for agentic applications, measuring how well agents retain and leverage information across prolonged task sequences. In both cases, the agent navigates a sequence of steps informed by all prior history, and evaluation centers on whether historical information is effectively integrated into next-step decisions. This paradigm inspires LongMedBench's clinical decision-making setting.

Table 1.2 Long-horizon and long-context benchmarks in the general domain. *Legend:* Long Context = requires processing contexts exceeding typical window sizes. Interactive = involves multi-step agent-environment interaction.

Benchmark	Multi-Turn	Long Context	Interactive	Reasoning	Key Design	Year
NarrativeQA [15]		✓			Full-book comprehension	2017
HotpotQA [16]				✓	Multi-doc reasoning	2018
RULER [14]		✓			Synthetic context stress test	2024
LoCoMo [17]	✓	✓	✓	✓	Multi-session dialogues	2024
LongMemEval [18]	✓	✓	✓	✓	Long-term chat memory	2024
TRIP-Bench [19]	✓	✓	✓	✓	Real-world interactive planning	2026
AMA-Bench [20]	✓	✓	✓	✓	Memory for agentic applications	2026

1.4 Memory-Augmented Architectures for Long-Horizon Interaction

To address the challenges exposed by long-horizon benchmarks, as shown in Table 1.3, a parallel line of research has developed memory-augmented architectures that extend LLM capabilities beyond their native context windows. The motivation is straightforward: an LLM's parameters are fixed at inference time, and its input context is finite. Without an external memory mechanism, information from prior documents, sessions, or patient visits either disappears or must be repeatedly reinserted into the prompt.

Retrieval-Augmented Generation (RAG) The foundational RAG paradigm [21] addresses this limitation by retrieving relevant documents from an external knowledge base at inference time and prepending them to the context window. In this formulation, retrieval functions as a read-only external memory: the model does not need to store all knowledge in its parameters or input prompt, but can condition generation on a small set of retrieved passages. This design has become a standard approach for factual grounding and domain adaptation, especially when the relevant knowledge base is large or frequently updated.

Agent Memory Systems However, long-horizon interaction requires more than one-shot retrieval from a static corpus. A personalized agent must preserve continuity across sessions, track evolving goals and states, update earlier assumptions, and selectively use prior experience when making future decisions [22]. MemGPT [23] introduced an OS-inspired memory hierarchy with working and archival storage, making memory management an explicit part of agent operation. Mem0 [24] provides a persistent, evolvable memory layer with add, search, and update operations. More recent frameworks push this idea further: A-MEM [25] models agentic memory decisions, where the agent autonomously determines what to retain; EverMemOS [26] proposes a self-organizing memory operating system for structured long-horizon reasoning; and Memory as Action [27] reframes memory management as an action the agent takes, integrating it into the decision loop rather than treating it as preprocessing. We adopt Mem0 [24] as the representative agent memory system in our experiments because it has achieved state-of-the-art performance on both the LoCoMo [17] and LongMemEval [18] datasets, and its clean API and production-ready design are well-suited for systematic ablation across memory granularities.

Table 1.3 Memory-augmented architectures for LLM agents. *Legend:* Dynamic = memory state evolves across interactions.

Framework	Strategy	Mechanism	Dynamic	Year	Innovation
RAG [21]	Retrieve-then-generate	Embed + Retrieve		2020	External knowledge access
MemGPT [23]	Hierarchical memory	Core + Archival paging	✓	2023	OS-inspired management
Mem0 [24]	Agent memory layer	Add + Search + Update	✓	2024	Multi-session persistence
A-MEM [25]	Agentic memory	Autonomous retention	✓	2025	Agent-driven memory
Memory as Action [27]	Memory-as-decision	Context curation loop	✓	2025	Memory in decision loop
EverMemOS [26]	Self-organizing OS	Structured long-horizon	✓	2026	Self-organizing memory

1.5 LongMedBench

LongMedBench is positioned at the intersection of the two fields surveyed above. It inherits the domain grounding of medical benchmarks while adopting the task design methodology of general long-horizon benchmarks.

1. **Innovation in medical AI:** LongMedBench is the first benchmark to combine real EHR data with interactive agent evaluation and systematic memory architecture ablation. No prior medical benchmark (Table 1.1) has systematically compared how different memory strategies—naive long-context, retrieval-augmented, and agent-managed memory—affect clinical decision-making.

2. **Innovation in long-horizon AI:** LongMedBench is the first benchmark to apply long-horizon interaction task design to real-world clinical data. Unlike general-domain benchmarks that use synthetic or curated text, LongMedBench tests long-horizon reasoning on noisy, real-world clinical trajectories with temporal structure and clinical consequence.

1.6 Research Questions

Specifically, focusing on Long-Horizon Decision-Making tasks within LongMedBench, this thesis investigates three questions:

1. **Memory granularity:** Does the format of historical information affect clinical decision quality?

2. **Memory architecture:** Do RAG and agent memory systems (Mem0) improve longitudinal decision-making relative to providing the full history in context?

3. **Historical depth:** Does decision performance plateau, improve, or degrade as more historical visits are included?

1.7 Contributions

Within the broader LongMedBench project, this thesis makes four contributions:

1. **EHR-to-Agent Data Pipeline.** A reproducible pipeline transforming raw MIMIC-IV [2] records into longitudinal event streams for interactive agent evaluation,

yielding 355 patients and 6,999 hospitalization visits (mean 19.72 visits/patient, 44.91 medical events/visit).

2. **Three-Tier Memory Architecture.** A memory framework operating at three granularities, enabling systematic ablation of memory effects on decision quality: Note Memory $\mathcal{M}_N^{(i,j)}$ (coarse-grained visit summaries), Event Memory $\mathcal{M}_E^{(i,j)}$ (fine-grained structured event sequences), and Context Memory $\mathcal{M}_C^{(j,T)}$ (immediate visit context).

3. **Long-Horizon Decision-Making Task.** An interactive evaluation protocol where the agent receives a partial patient trajectory through controlled memory and must predict the next clinical action from a predefined action space, with ground truth extracted from actual clinical records.

4. **Empirical Finding.** Through experiments across GPT-5-mini [28], DeepSeek-v3.2 [29], and Qwen-Turbo [30] comparing Naive Long-Context, RAG, and Mem0 strategies, we find that memory augmentation provides no significant improvement for longitudinal clinical decision-making. Decision accuracy depends primarily on immediate context, revealing a retrieval-reasoning gap that challenges current memory-centric approaches.

1.8 Scope and Structure

This thesis focuses exclusively on Long-Horizon Decision-Making Tasks on LongMedBench. The Fact-based QA and Temporal Reasoning tasks are covered in collaborators' separate theses.

Chapter 2 presents the data processing pipeline, three-tier memory architecture, decision-making task design, experimental setup, and results. Chapter 3 discusses the implications of our findings, acknowledges limitations, and outlines future research directions.

Chapter 2. Methodology and Experiments

This chapter presents the LongMedBench decision-making pipeline: from raw EHR data to interactive agent evaluation. Section 2.1 describes the data processing pipeline. Section 2.2 presents the three-tier memory architecture. Section 2.3 formulates the decision-making task. Section 2.4 details the experimental design. Section 2.5 presents results and analysis.

2.1 EHR Data Processing Pipeline

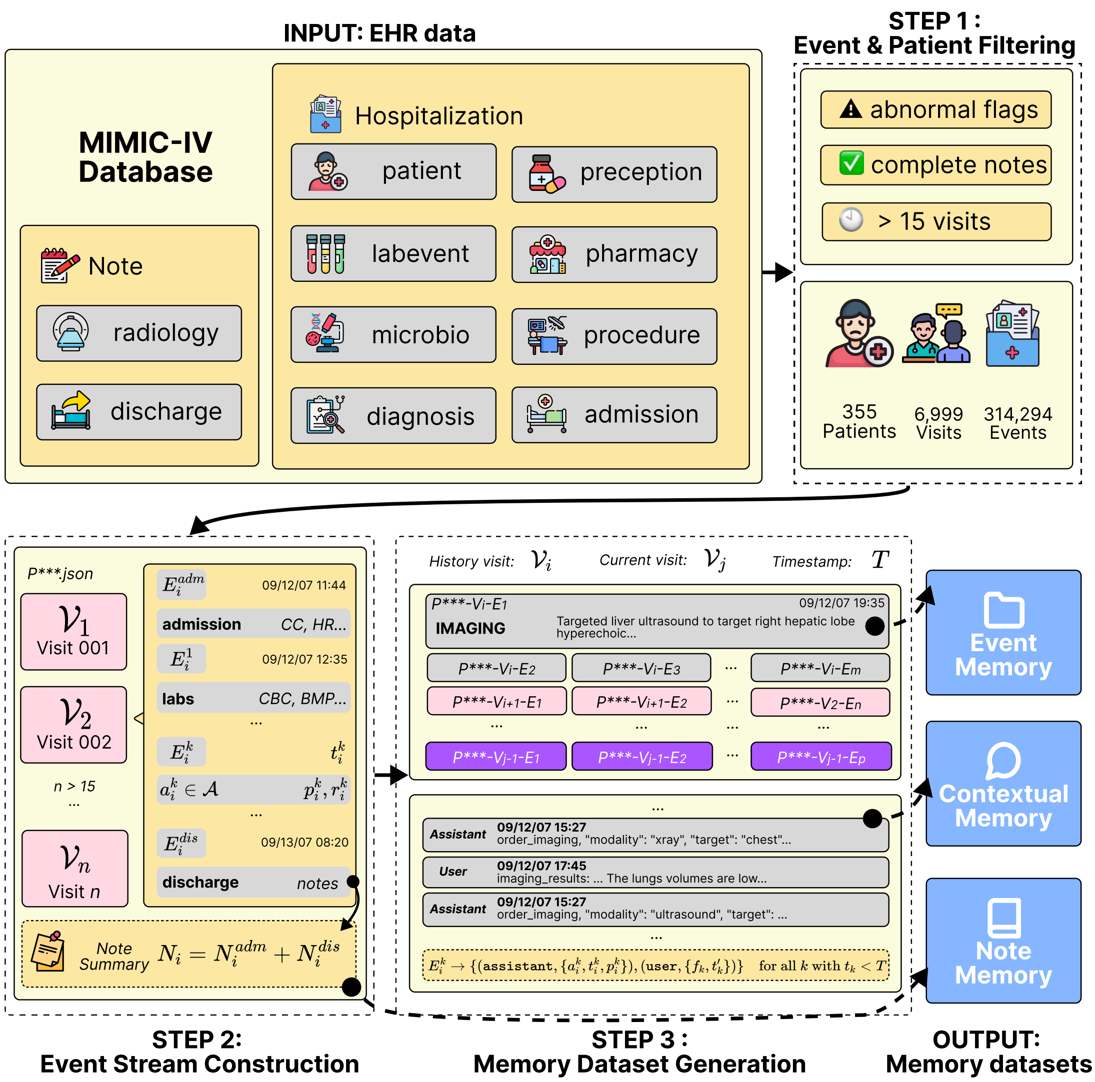

Figure 2.1 Overview of the EHR data processing pipeline.

LongMedBench is constructed from MIMIC-IV v3.1 [2], a publicly available deidentified database containing comprehensive clinical data for over 100,000 patients at Beth Israel Deaconess Medical Center (2008–2022). To convert this database into an interactive agent-compatible environment, we design a three-stage pipeline (Figure 2.1).

2.1.1 Data Preprocessing & Event Filtering

Event-Level Filtering. All structured clinical events are extracted from the `hosp` module, including laboratory measurements, microbiology cultures, medication prescriptions, pharmacy bills, diagnoses and procedures. Among these, we retain only laboratory results flagged as abnormal; normal lab values are excluded. Other events are retained in full. The rationale for this selective filtering strategy is discussed in Chapter 3.

From the `admissions` table, we decompose each hospitalization into two boundary events. The admission event captures the patient's demographics, chief complaint, and initial physical examination findings at intake. The discharge eventcaptures the discharge narrative and the final diagnoses recorded in `diagnoses_icd`.

Patient-Level Preprocessing. We require each hospitalization to have a complete discharge summary from the `note` module whose timestamp aligns with the corresponding admission record in `hosp/admissions`; visits lacking a matched discharge note are excluded. From the remaining visits, we select patients with at least 15 recorded hospitalizations. The rationale for these filtering decisions is discussed in Chapter 3.

Note Preprocessing. Radiology reports are converted to structured imaging events via regular expression parsing: the modality *(e.g., CT, X-ray, MRI)* and body region *(e.g., chest, abdomen)* are extracted as the action type and parameter, respectively, while the full report text is retained as the observation result. Discharge notes N_i are decomposed into admission information N_i^{adm} and discharge information N_i^{dis}, then attached to the corresponding admission or discharge events. Both summaries are preserved in narrative format for use in Note Memory (Section 2.2.1) and ground-truth extraction.

Resulting Cohort. The filtered dataset comprises 355 patients with 6,999 inpatient visits. Mean visits per patient: 19.72 (median = 18.00, SD = 5.72). Mean medical events per visit: 44.91 (median = 25.00, SD = 70.32). The distribution of data is shown in Figure 2.2. This dense multi-session structure challenges models on cross-visit memory retention and temporal reasoning.

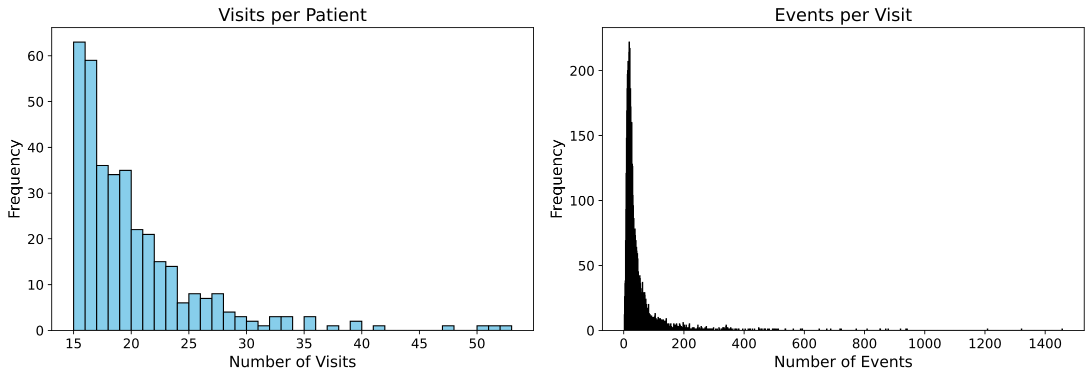

Figure 2.2 Distribution of visits per patient in the filtered cohort.

2.1.2 Event Stream Construction

We transform filtered records into a unified, time-ordered event stream for each patient. A medical event is any discrete clinical occurrence extracted from the structured hospitalization data. Formally, the k_{th} event in $\mathcal{V}_i$ is $E_i^k := \{a_i^k, t_i^k, p_i^k, o_i^k\}$, where a_i^k is the action type from the clinical action space $\mathcal{A}$, *including imaging, lab tests, medication, etc.*, t_i^k is the timestamp, p_i^k is the action parameter *(e.g., CT, X-ray, etc. for imaging)*, and o_i^k is the observation result *(e.g., a radiology report)*.

We define a visit as the period spanning from an admission event E_i^{adm} to the corresponding discharge event E_i^{dis}. All other structured events occurring between these two timestamps are collected and ordered chronologically. Formally, each visit $\mathcal{V}_i$ can be represented as $\mathcal{V}_i = \{E_i^{adm}, E_i^1, E_i^2, \ldots, E_i^m, E_i^{dis}\}$. The complete patient stream can therefore be denoted as $\mathcal{S}_{\mathcal{P}} = \{\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_n\}$, where n is the total number of visits.

2.2 Three-Tier Memory Architecture

A central design principle is the controlled provision of historical clinical information. We define a three-tier memory architecture that systematically varies both the granularity of historical information. Given the earliest historical visit $\mathcal{V}_i$, the current visit $\mathcal{V}_j$, and the current timestamp T, the agent's memory access is strictly bounded to prevent event leakage.

2.2.1 Note Memory (Coarse-Grained)

$\mathcal{M}_N^{(i,j)} := \{N_i, N_{i+1}, \ldots, N_{j-1}\}$, which contains continuous visit-level discharge summaries from $\mathcal{V}_i$ to $\mathcal{V}_{j-1}$. Note Memory is compact and preserves clinical narrative structure, emphasizing the global trajectory, but discards fine-grained temporal and event-level details.

2.2.2 Event Memory (Fine-Grained)

$\mathcal{M}_E^{(i,j)} := \bigcup_{k=i}^{j-1} \mathcal{V}_k$, which contains all clinical events from $\mathcal{V}_i$ to $\mathcal{V}_{j-1}$. Events are flattened into a single chronological sequence, discarding visit-level boundaries and forming a unified patient trajectory. Event Memory is information-complete, but its volume (mean 900 events across 20 visits) approaches or exceeds typical LLM context windows, creating a natural stress test for both long-context capabilities and retrieval strategies.

2.2.3 Context Memory (Immediate)

$\mathcal{M}_C^{(j,T)} := \{(r_0, m_0), (r_1, m_1), \ldots, (r_P, m_P)\}$, where r_k and m_k denote the dialog role and message content, respectively. Any event E_j^k in $\mathcal{V}_j$ with timestamp $t_j^k < T$ is transformed into an LLM-style conversation trajectory, simulating the agent's instant interaction state up to the decision point. Context Memory represents the baseline information always available to the agent. Historical memory ($\mathcal{M}_N^{(i,j)}$ or $\mathcal{M}_E^{(i,j)}$) is the experimental variable, enabling isolation of the marginal contribution of history beyond the immediate clinical context.

2.2.4 Memory Provision Strategies

Given the fixed memory modules $\{\mathcal{M}_N^{(i,j)}, \mathcal{M}_E^{(i,j)}, \mathcal{M}_C^{(j,T)}\}$, three strategies control how historical memory is provided to the agent:

Naive Long-Context (Baseline). $\mathcal{M}_N^{(i,j)}$ or $\mathcal{M}_E^{(i,j)}$ is provided in full within the model's context window alongside $\mathcal{M}_C^{(j,T)}$, testing native long-context attention without external retrieval assistance.

RAG. $\mathcal{M}_N^{(i,j)}$ or $\mathcal{M}_E^{(i,j)}$ is externally indexed. At decision time, $\mathcal{M}_C^{(j,T)}$ serves as the retrieval query; top-k relevant chunks are retrieved and concatenated with the context. This tests whether selective retrieval overcomes lost-in-the-middle effects.

Mem0 (Agent Memory). We integrate the Mem0 framework [24], providing a persistent, evolvable memory store. The agent can add, search, and update memory across a patient's trajectory, actively managing what is stored and retrieved. This tests whether agentic memory management outperforms passive retrieval.

2.3 Long-Horizon Decision-Making Task (T3-D)

The T3-D task evaluates an LLM agent's ability to make proactive clinical decisions based on partial patient trajectories. Unlike static benchmarks that present complete histories for classification, T3-D simulates interactive decision-making: the agent receives information incrementally and must determine the next action at each decision point.

2.3.1 Task Formulation

At each decision point, the agent receives: (1) a system prompt establishing clinical role and task instructions; (2) Context Memory $\mathcal{M}_C^{(j,T)}$; (3) Historical Memory ($\mathcal{M}_N^{(i,j)}$ or $\mathcal{M}_E^{(i,j)}$, at controlled granularity and provision strategy); and (4) a decision query directing the agent to predict the next clinical action. The agent outputs a predicted action $\hat{a}$ from the clinical action space $\mathcal{A}$, which is compared against ground-truth action a^* extracted from actual records.

2.3.2 Action Space and Ground Truth

The action space $\mathcal{A}$ includes: Imaging (CT, X-ray, MRI, Ultrasound), Laboratory Tests (CBC, metabolic panels, cultures), Medications (drug prescriptions with name/dose/route), Procedures (intubation, central lines), Consultations (specialty referrals), Admission/Discharge actions, and Transfers (between care units).

Ground-truth actions are extracted through structured field extraction from MIMIC-IV tables and regular expression parsing of discharge notes. Extracted actions are validated for temporal coherence (e.g., medication order before administration, discharge terminating the visit).

Decision points are sampled to cover diverse clinical scenarios: early in the visit (post-admission), mid-visit (during active treatment), and late in the visit (near discharge). Each patient contributes multiple decision points across their visits, yielding approximately 9,600 decision instances.

2.3.3 Evaluation Metrics

Exact Match (EM): Proportion of instances where predicted action matches ground truth exactly (action type + parameter). **Hit-Any (HA):** Proportion where predicted action type matches ground truth type, regardless of parameter (e.g., "CT Chest" vs. "X-ray Chest" both count as imaging). Per-category metrics are reported disaggregated by action type.

These metrics capture complementary aspects: EM is strict, penalizing clinically reasonable alternatives; HA relaxes parameter specificity, acknowledging that action category may be more informative.

2.4 Experimental Setup

2.4.1 Models

We evaluate three state-of-the-art LLMs: GPT-5-mini (400K context window), DeepSeek-v3.2 (128K context, with and without chain-of-thought reasoning mode), and Qwen-Turbo (128K context, baseline for ablation studies). All are evaluated through their respective APIs with standardized prompts.

2.4.2 Experimental Conditions

We design a factorial ablation varying four factors:

1. **Memory Granularity:** Note Memory vs. Event Memory.

2. **Provision Strategy:** Naive Long-Context vs. RAG vs. Mem0.

3. **Historical Depth (n):** Number of injected historical visits: $n \in \{0, 1, 2, 3, \infty\}$. $n = 0$ provides Context Memory $\mathcal{M}_C^{(j,T)}$ only (no history) as baseline.

4. **Reasoning Mode:** Standard vs. chain-of-thought (DeepSeek-v3.2 only).

Qwen-Turbo serves as the primary ablation vehicle; key conditions are replicated on GPT-5-mini and DeepSeek-v3.2 for cross-model validation.

2.5 Results and Analysis

2.5.1 Overall Decision-Making Performance

Table 2.1 summarizes T3-D performance across models. Decision-making accuracy falls within a narrow band of 0.55–0.62 (Exact Match), confirming that T3-D presents a genuine challenge that current LLMs have not mastered. The modest ceiling validates the benchmark's difficulty and its capacity to differentiate future improvements.

Table 2.1 Overall T3-D performance (Exact Match).

Model	Note Mem.	Event Mem.	Context
GPT-5-mini	[TODO]	0.607	400K
DeepSeek-v3.2	[TODO]	0.607	128K
DeepSeek-v3.2-thinking	[TODO]	0.593	128K
Qwen-Turbo	[TODO]	0.604	128K

2.5.2 Memory Granularity: Note vs. Event

Table 2.2 compares performance across memory granularities. Structured Event Memory offers slight advantages over Note Memory for most models, but the difference is modest—consistent with the interpretation that for proactive clinical planning, immediate context outweighs historical granularity.

Table 2.2 Effect of memory granularity on T3-D (Naive Long-Context).

Model	Note	Event	Δ
GPT-5-mini	[TODO]	0.607	[TODO]
DeepSeek-v3.2	[TODO]	0.607	[TODO]
Qwen-Turbo	[TODO]	0.604	[TODO]

2.5.3 Memory Architecture: Naive vs. RAG vs. Mem0

This is the central experimental question. Table 2.3 presents the memory architecture ablation using Qwen-Turbo.

Table 2.3 Effect of memory architecture on T3-D (Qwen-Turbo, Event Memory).

Architecture	T3-D Acc.	Δ vs. Naive	Notes
Naive (Full)	[TODO]	—	Baseline
RAG	[TODO]	[TODO]	Top-k retrieval
Mem0	[TODO]	[TODO]	Agent memory

Finding. Neither RAG retrieval nor agent memory management (Mem0) provides significant improvement over the naive long-context baseline. In some conditions, augmentation slightly degrades performance. This negative result is the central empirical contribution: it reveals that the bottleneck in longitudinal clinical decision-making is not retrieval (locating relevant history) but integration (understanding how historical information bears on the current decision). Current architectures optimize for retrieval quality, but our results indicate that for clinical planning, closing the gap between retrieval and reasoning is the critical challenge.

2.5.4 Historical Depth: Visit Injection Ablation

Table 2.4 shows performance as a function of injected historical visits (n). The $n = 0$ baseline ($\mathcal{M}_C^{(j,T)}$ only, no history) is particularly informative.

Finding. Performance does not monotonically improve with historical depth. The $n = 0$ baseline achieves accuracy comparable to full-history conditions, indicating that decisions are driven primarily by the immediate clinical presentation. Adding history

Table 2.4 Effect of historical depth on T3-D (Qwen-Turbo, Event Memory).

n (Visits)	T3-D Acc.	Interpretation
0 (Context only)	[TODO]	Baseline: no history
1	[TODO]	Single prior visit
2	[TODO]	Two prior visits
3	[TODO]	Three prior visits
∞ (All)	[TODO]	Complete history

introduces noise that can offset the value of relevant historical information—consistent with the lost-in-the-middle effect [31].

2.5.5 Chain-of-Thought Reasoning

For DeepSeek-v3.2, explicit chain-of-thought reasoning reduces T3-D accuracy from 0.607 to 0.593. Qualitative analysis of reasoning traces suggests that thinking mode makes the model more conservative—it identifies information gaps and hesitates rather than committing to a decision. This highlights a tension between clinical caution and benchmark performance.

2.5.6 Cross-Model Comparison

[TODO: Brief comparison of performance across GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo. Note whether larger context windows (400K vs. 128K) translate to better decision-making, and whether model architecture matters more than context capacity.]

2.5.7 Error Analysis

[TODO: Summary of error patterns—

- Action category confusion matrix (which types are most often confused?)
- Temporal sensitivity (does accuracy degrade deeper into a visit or later in a trajectory?)
- Patient complexity correlation (does performance correlate with visit count or event density?)
- Clinically reasonable alternatives (among incorrect EM predictions, what fraction represent clinically defensible actions?)

]

2.5.8 Summary of Findings

The experimental results converge on a coherent narrative: (1) Clinical decision-making remains challenging (accuracy 0.55–0.62 across all models); (2) Memory granularity matters modestly (Event slightly outperforms Note); (3) Memory augmentation (RAG, Mem0) does not significantly improve decision-making—the central negative result; (4) Immediate context dominates historical context; (5) Chain-of-thought reasoning can degrade performance through increased conservatism. These findings collectively reveal a retrieval-reasoning gap whose implications we explore in Chapter 3.

2.5.9 Qualitative Case Study

To illustrate the decision-making task concretely, consider a representative patient from our cohort: a 68-year-old male with chronic kidney disease, type 2 diabetes, and hypertension, with 21 hospitalizations over five years. At a mid-visit decision point during his 15th admission (presenting with fever and elevated creatinine), the agent receives:

- **Context Memory ($\mathcal{M}_C^{(j,T)}$):** Admission event (fever, hypotension), initial lab results (elevated WBC, creatinine 2.8 mg/dL), and one prior imaging order (chest X-ray, pending result).
- **Historical Memory (Event):** Events showing that during prior febrile episodes, the patient responded to vancomycin but developed acute kidney injury requiring dose adjustment; a CT abdomen from visit 12 revealed nephrolithiasis.
- **Decision Query:** "Based on the patient's history and current status, what should be the next clinical action?"

The ground-truth action (from actual records) is ordering blood cultures and starting empiric vancomycin at renally adjusted dosing. An agent relying solely on $\mathcal{M}_C^{(j,T)}$ might order broad-spectrum antibiotics at standard dosing; an agent effectively integrating history would recognize the need for renal adjustment based on prior vancomycin-associated AKI and the current elevated creatinine. This case illustrates the type of decision where historical integration should change the answer—and where our benchmark can measure whether it does.

2.5.10 Discussion of Negative Results

The finding that memory augmentation does not improve decision-making may appear discouraging, but it carries productive implications. First, it identifies a genuine capability gap that current architectures cannot bridge through scale or retrieval quality alone—the gap is architectural, not parametric. Second, it provides a clear target for future work: architectures that explicitly model the relationship between retrieved history and current decisions, rather than treating retrieval as an independent preprocessing step. Third, it validates LongMedBench's sensitivity: a benchmark that showed all approaches performing equally well would be uninformative; one that reveals systematic failure modes provides actionable diagnostic signal.

The contrast with EHR-Bench [12] is instructive. While EHR-Bench demonstrated that LLMs can predict next clinical events from complete histories with non-trivial accuracy, our agent-based setting reveals that this capability does not translate to interactive decision-making where history must be selectively retrieved and integrated. The two paradigms evaluate complementary but distinct capabilities—static prediction vs. interactive planning—and gap between them highlights the importance of evaluation methodology in assessing clinical AI readiness.

Chapter 3. Discussion and Conclusion

3.1 Summary of Findings

The experimental results presented in Chapter 2 yield five key findings:

1. Decision-making remains challenging. Across GPT-5-mini, DeepSeek-v3.2, and Qwen-Turbo, T3-D accuracy ranges from 0.55–0.62 (Exact Match), substantially below clinical deployment requirements. The consistency of this ceiling across architectures suggests a fundamental limitation of current LLM approaches rather than model-specific weakness.

2. Memory granularity matters modestly. Structured Event Memory slightly outperforms Note Memory in most conditions, but the effect is small relative to the overall performance ceiling. For proactive clinical planning, immediate context dominates.

3. Memory augmentation does not improve decision-making. This is the central finding. Neither RAG nor Mem0 provides significant improvement over the naive long-context baseline. In some conditions, augmentation slightly degrades performance.

4. Immediate context dominates historical context. The $n = 0$ baseline ($\mathcal{M}_C^{(j,T)}$ only, no history) achieves accuracy comparable to full-history conditions. Adding historical visits does not monotonically improve decisions.

5. Chain-of-thought can degrade performance. For DeepSeek-v3.2, reasoning mode reduces accuracy from 0.607 to 0.593, likely because explicit reasoning induces conservatism that penalizes decisive action under exact-match evaluation.

3.2 The Retrieval-Reasoning Gap

These findings collectively reveal a **retrieval-reasoning gap**: a disconnect between LLMs' ability to locate relevant historical information and their ability to integrate that information into proactive clinical planning. This explains why RAG and Mem0 fail: these architectures optimize retrieval quality, but for clinical decision-making, retrieval is not the bottleneck. The model can access relevant history (either in full context or via retrieval), but cannot effectively use it to inform decisions.

We hypothesize three contributing mechanisms. First, clinical reasoning requires causal, not associative, inference—determining whether a prior treatment was effective requires understanding why it was given and how the patient responded, not just retrieving that it occurred. Second, the signal-to-noise ratio in longitudinal EHR data is low: a patient's 900 events contain only a fraction relevant to any single decision, overwhelming attention mechanisms. Third, current LLM training data (web text, code, books) lacks the temporal-causal structure of clinical trajectories, creating a distribution shift.

3.3 Implications

3.3.1 For Memory Architecture Design

Current memory architectures (RAG, Mem0, MemGPT) operate on the assumption that better retrieval yields better performance. Our findings challenge this assumption for clinical decision-making. Future architectures should emphasize reasoning over retrieved information: temporal attention mechanisms that encode causal relevance, graph-based representations that capture clinical relationships, and structured reasoning layers that bridge retrieval and decision.

3.3.2 For Benchmark Design

If decision-making performance is dominated by immediate context, benchmarks that primarily test retrieval may overestimate clinical readiness. We advocate for adversarial task design—constructing decision instances where history changes the correct answer—to prevent models from succeeding through immediate-context heuristics alone. Additionally, distinguishing the subset of decisions where history genuinely matters from those driven by current presentation is critical for targeted evaluation.

3.3.3 For Clinical AI Deployment

The finding that immediate context dominates is not necessarily a weakness—in many clinical scenarios, the current presentation is indeed the most important determinant. The challenge lies in identifying decisions where ignoring history constitutes clinical error (e.g., prescribing a medication that previously caused an adverse reaction) and ensuring the agent has the architecture to recognize and act on such cases.

3.4 Limitations

Data Scope. LongMedBench is constructed from MIMIC-IV, a single-institution database. Clinical practice patterns and documentation conventions vary across institutions, and findings may not generalize without cross-institutional validation.

Cohort Bias. Several design decisions in our data pipeline shape the cohort composition. At the event level, we retain only laboratory results flagged as abnormal, following Hager et al. [32], who demonstrated that presenting models with only abnormal lab values improved diagnostic performance in MIMIC-CDM by reducing noise from clinically uninformative normal results. This filtering makes the evaluation more clinically realistic—like human clinicians, the agent reasons from salient pathological signals—but may underrepresent the diagnostic challenge of distinguishing normal variation from true abnormality. At the patient level, we require each hospitalization to have a matched discharge note from the `note` module and select patients with ≥ 15 hospitalizations. We chose this threshold after examining the trade-off between trajectory length and cohort size: lowering to ≥ 10 visits yields trajectories too short to meaningfully evaluate cross-visit reasoning; raising to ≥ 20 visits produces trajectories of over 20 steps but retains only approximately 100 patients, limiting statistical power. At ≥ 15 visits, the resulting cohort of 355 patients compares favorably to sample sizes in other EHR-based agent benchmarks—MedAlign enrolled 276 patients [5] and DiReCT enrolled 511 patients [33]. Nevertheless, this threshold selects for high-acuity, chronic, multi-morbid patients—the population for whom longitudinal reasoning matters most—and findings may not generalize to lower-acuity settings or the broader patient base.

Task Simplification. T3-D reduces clinical decision-making to action-type prediction within a predefined action space. Real decisions involve drug dosing, contraindication checking, patient preferences, resource availability, and guideline compliance not captured by our formulation.

Evaluation Rigor. Exact Match penalizes clinically reasonable alternatives. While

Hit-Any partially addresses this, neither metric replaces clinical expert judgment. We recommend human clinician evaluation as a future validation step.

Model Coverage. Our evaluation covers three general-purpose LLMs. Models specifically trained for clinical tasks (e.g., EHR-R1, Med-PaLM) may exhibit different patterns not captured here.

3.5 Future Work

Task Refinement. Adversarial decision instances where history changes the correct answer would directly test temporal integration. Counterfactual evaluation—altering history and measuring decision sensitivity—would isolate causal effects.

Memory Architecture Development. Temporal graph memory, selective history summarization (compressing irrelevant while preserving salient information), and memory-aware fine-tuning on synthetic longitudinal trajectories are promising directions.

Human Baseline. A clinician evaluation study contextualizing the 0.55–0.62 accuracy range would validate benchmark difficulty and enable calibrated assessment of model progress.

Cross-Institutional Validation. Extending the pipeline to eICU, HiRID [34], or STARR would test generalizability across institutions and patient populations.

Multimodal Extension. Incorporating medical imaging and physiological waveforms would increase clinical fidelity as vision-language and multimodal LLMs mature.

Outcome Correlation. Linking agent decisions to simulated patient outcomes (mortality, readmission, length of stay) would provide the ultimate validation of decision-making quality.

3.6 Conclusion

Clinical practice is inherently longitudinal—patients accumulate histories over years, and clinicians must integrate evidence across this temporal expanse. As LLMs move toward clinical deployment, their capacity for longitudinal reasoning becomes a critical benchmark.

This thesis presented the decision-making component of LongMedBench, investigating how contemporary LLMs use historical clinical information to inform proactive planning through a three-tier memory architecture and systematic ablation experiments. The central finding is that memory augmentation—the dominant paradigm for extending LLM capabilities—does not significantly improve longitudinal clinical decision-making. Decision accuracy depends primarily on immediate context, and historical information, however organized, contributes marginal additional value.

This finding clarifies the nature of the challenge. The bottleneck is not retrieval—finding the right facts—but integration—understanding how those facts collectively inform decisions. Closing this retrieval-reasoning gap requires architectures that go beyond retrieval to support structured reasoning over temporal clinical evidence. The path to clinically capable AI agents is longer than single-encounter benchmarks suggest, but by identifying where current approaches fall short, LongMedBench provides a platform for measuring and ultimately closing this gap.

3.6.1 Relationship to the Broader LongMedBench Project

While this thesis focuses on T3-D (Decision-Making), placing our findings in the context of the broader LongMedBench project reveals complementary insights. The T3-A (Fact-based QA) evaluation found that RAG and agent memory systems significantly improve factual retrieval from long clinical histories [24]. T3-N (Temporal Reasoning) revealed that LLMs struggle with implicit temporal inference (visit ordering without explicit timestamps) despite strong performance when timestamps are provided.

The contrast between these task suites sharpens our central finding: memory augmentation does help for retrieval tasks (T3-A), but does not help for decision-making tasks (T3-D). This dissociation confirms that the retrieval-reasoning gap is task-specific—it manifests when the agent must do more than locate facts, when it must plan proactively. This insight would not be visible from any single task suite; it emerges only from the systematic, multi-dimensional evaluation that LongMedBench provides.

3.6.2 Practical Considerations for Reproducibility

The MIMIC-IV data itself requires credentialed access through PhysioNet. To facilitate reproducibility and future extensions, we will release the data processing pipeline, task generation logic, and evaluation harness, if LongMedBench is admitted by MICCAI 2026.

API-based model evaluation used default settings and fixed random seeds where applicable. Full prompt templates, including system prompts, memory formatting conventions, and decision queries, are documented in the released codebase to ensure exact replication.

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Appendix

(if applicable)

You can attach your publication, patent document, etc here.

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(if applicable)

(Main text: Times New Roman, size 12, 1.5 linespace)

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